

First Animal/Bird Language Translator

Enabling You To Talk With Nature

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All living beings, be they flora or fauna, have a unique way of communicating and expressing their emotions using various sounds and signals. We humans are blessed with the capability to observe, learn, and speak many languages. We also possess an innate curiosity to know more about our ecosystem and the world around us. It is our wish to understand nature's feelings. At the same time, we also want to share our feelings.

This made me think, why not observe, research, and collect sounds of animals and birds, extract the emotions in those sounds, and make a device that translates their language to ours and vice-versa? That would be amazing! Based on research by scientists, there are several open

source datasets regarding nature's communication methods available for use by anyone.

So, by employing these open source datasets you can develop and train a machine language (ML) model that understands the emotions in different animal sounds and classifies them accordingly. You will also be able to deploy the ML model for translating human language into nature's language and back, enabling efficient communication.

Doesn't it sound awesome? So, without wasting any time, let us start our beautiful journey.

Preparing datasets

The ML needs to be fed with correct data regarding sounds and emotions. You can download various open source datasets of different birds and animals available online such as sounds made by an elephant for communicating motion, love, care, anger, etc.

After downloading such animal sounds, compile them into datasets to train the ML model. You

can use tools such as Tensorflow, Edge Impulse, SensiML, Teachable Machine, and many others available for this purpose. Here, I am using Edge Impulse.

Open the Raspberry Pi terminal and install Edge Impulse dependencies. Then using the following commands, create a new project named Fauna Translator:

```
curl -sL https://deb.nodesource.com/
setup_12.x | sudo bash -

sudo apt install -y gcc g++ make build-
essential nodejs sox gstreamer1.0-tools
gstreamer1.0-plugins-good gstreamer1.0-
plugins-base gstreamer1.0-plugins-
base-apps

npm config set user root && sudo npm install
edge-impulse-linux -g --unsafe-perm
```

After this, connect the Raspberry Pi project with Edge Impulse using Edge-impulse-linux

Next, open the terminal and select the project name, after which you are given a URL for feeding the datasets. In that URL, enter the captured sounds of animals and appropriately label

Components		Table 1
Component name	Quantity	Description
Raspberry Pi 4	1	2GB
AIY Voice Bonnet	1	Sound data processing



Pictures of author's prototype

